Winter Training



CYBER SECURITY

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CYBER WINTER

- WEEK 1 - INTRODUCTION TO CYBERSECURITY AND ETHICAL HACKING
- WEEK 2 - NETWORKING FUNDAMENTALS FOR ETHICAL HACKING
- WEEK 3 - INTRODUCTION TO FOOTPRINTING AND RECONNAISSANCE
- WEEK 4 - SCANNING AND ENUMERATION TECHNIQUES
- WEEK 5 - SYSTEM AND NETWORK EXPLOITATION BASICS WITH WIRELESS ATTACKS
- WEEK 6 - WEB APPLICATION SECURITY FUNDAMENTALS , REMEDIATIONS AND REPORTING

ABOUT CYBER SECURITY



Cybersecurity is all about keeping our digital world safe. It means protecting computers, networks, and data from hackers, viruses, and all sorts of cyber threats. Think of it like locking the doors and windows of your home, but in this case, it's to keep our online information safe.

Why is Cybersecurity Important?

Cybersecurity isn't just for big companies—it affects everyone. Here's why it matters:

- Personal Protection: Without cybersecurity, personal information (like your bank details, private messages, or photos) could end up in the wrong hands.
- Business Safety: For companies, a cyber attack could mean a huge financial loss, damage to their reputation, and loss of trust from their customers.
- National Security: At a larger scale, cybersecurity helps protect a country's critical systems, like power grids, healthcare, and defense.

Cyber attacks are only becoming more common, so strong cybersecurity is essential to keep both people and businesses safe in today's tech-driven world.

CIA TRIAD

Confidentiality
This is all about privacy. Only people who are supposed to see the information should be able to access it. Imagine your private messages, if someone else could read them, that's a breach of confidentiality.

Example: Imagine a hacker breaks into a company's database and steals customer credit card information. This is a classic confidentiality breach.

Integrity
This means making sure the information we access or share hasn't been altered without permission. If a hacker changes your medical records, that's a serious breach of integrity.

Example: If someone hacks into a bank and alters account balances, it impacts the integrity of their data.

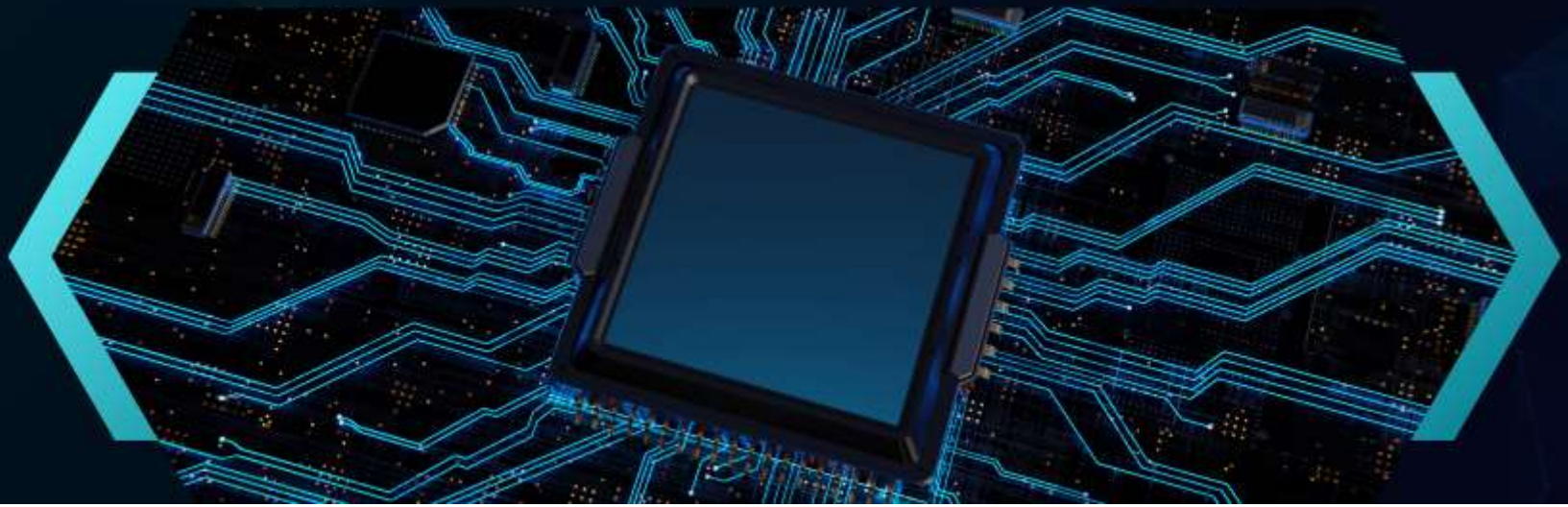
Availability
This is about making sure information is accessible when needed. If a website is down due to an attack and customers can't access it, that's a failure of availability.

Example: A denial-of-service (DoS) attack can crash a website, making it unavailable to customers.



WHY YOU SHOULD CARE ABOUT CYBERSECURITY

We live in a connected world, where our personal and professional lives are closely tied to technology. Cybersecurity helps protect not just our data, but our trust in the systems we rely on daily. With each new tech advancement, there's a new risk, but with good cybersecurity practices, we can stay one step ahead and keep ourselves and our information safe.



Winter Training

INTRODUCTION TO ETHICAL HACKING



What is Ethical Hacking

Ethical hacking is the practice of legally breaking into systems to find and fix security issues before malicious hackers can exploit them. Think of ethical hackers as the "good guys" as they use their skills to protect rather than harm. Companies hire them to identify weaknesses and secure their systems from real threats.

Why do we need Ethical Hackers?

In the digital world, new threats pop up constantly. Even with the best cybersecurity systems in place, there can still be hidden vulnerabilities. Ethical hackers help companies:

01 **Stay Ahead of Cybercriminals:** By finding weaknesses first, ethical hackers prevent attacks before they happen.

02 **Protect Sensitive Data:** From financial records to personal information, ethical hackers help safeguard important data.

03 **Maintain Trust:** Companies that take cybersecurity seriously gain trust from their customers and avoid reputational damage from data breaches.

06

Legal and Ethical Consideration

Ethical hacking isn't just about having technical skills; it also comes with big responsibilities:

01 **Permission is Key:** Ethical hackers only work with permission from the company or organization they're testing. Unauthorized hacking, even with good intentions, is illegal.

02 **Responsible Disclosure:** When ethical hackers find a vulnerability, they're responsible for reporting it directly to the company, not sharing it with the public or selling it.

03 **Professional Conduct:** Ethical hackers follow a code of ethics to make sure their work is in the best interest of the client and society.

The Role of an Ethical Hacker

Ethical hackers work in various roles and industries. They may:

01 **Penetration Test:** Try to break into systems to identify and fix weaknesses.

02 **Vulnerability Assessments:** Review systems for potential risks without actively exploiting them.

03 **Security Training:** Help train employees to recognize and avoid security threats, like phishing scams.

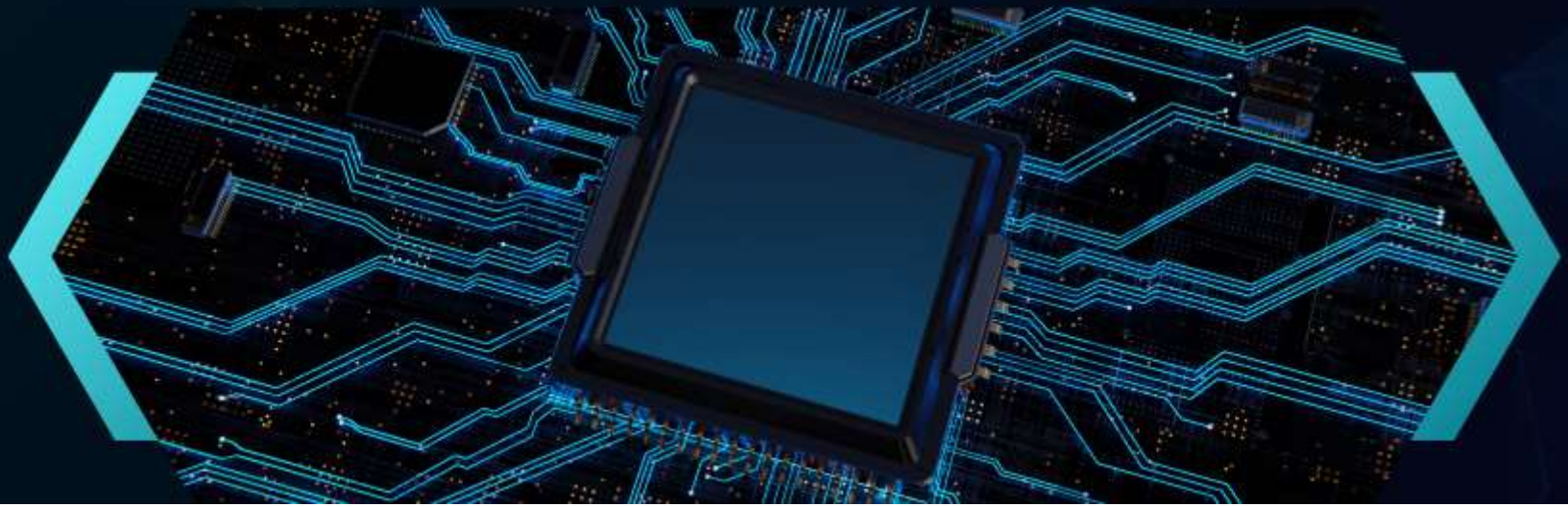


HOW ETHICAL HACKING DIFFERS FROM CRIMINAL HACKING

Ethical hacking and criminal hacking use similar skills, but the intentions are worlds apart:

01 **Ethical Hackers:** Work with permission to protect systems, improve security, and report weaknesses responsibly.

02 **Criminal Hackers:** Break into systems without permission, often to steal data, cause harm, or make a profit illegally.



STARTING GOAL AND ENDING GOAL OF WINTER TRAINING







Our initial goal is to make sure everyone has a strong understanding of cybersecurity basics.

STARTING GOAL : LAYING THE FOUNDATION





- **Understanding Key Concepts:** We'll start by defining cybersecurity, ethical hacking, and the important terms you'll encounter. Think of this as building a vocabulary to help you understand the world of cybersecurity.
- **Recognizing Threats and Risks:** Early on, you'll learn about common threats (like malware and phishing) and the risks these pose to individuals and organizations.
- **Setting Up a Lab Environment:** We'll guide you in setting up a secure virtual environment, which will act as your "sandbox" for safe practice without risking your own system.

ENDING GOAL : BUILDING PRACTICAL SKILLS

- **Identify Vulnerabilities:** You'll be able to spot weaknesses in systems, just like ethical hackers do.
- **Use Common Tools:** You'll be familiar with tools like Kali Linux, Metasploit, and other ethical hacking programs. These tools are essential for penetration testing and vulnerability assessments.
- **Understand Legal and Ethical Boundaries:** You'll know how to responsibly report vulnerabilities and understand the legal considerations of ethical hacking.
- **Perform Basic Penetration Testing:** Finally, you'll have the skills to run basic penetration tests in a lab setting, allowing you to test systems and learn how to protect them.





KEY TERMINOLOGY IN CYBERSECURITY

- **Access Control:** Techniques that restrict unauthorized access to systems and data. Types include Discretionary Access Control (DAC), Mandatory Access Control (MAC), and Role-Based Access Control (RBAC).
- **Authentication:** The process of verifying the identity of a user or device. Common methods include passwords, biometrics, and multi-factor authentication (MFA).
- **Authorization:** Grants permissions to a user or system to access resources based on their authenticated identity.
- **Adware:** Software that automatically displays or downloads ads, typically without user consent. Often used as a revenue-generating method by malicious programs.
- **Backdoor:** A hidden method of bypassing normal authentication, allowing remote access to a system. Often installed by attackers for persistent access.
- **Brute Force Attack:** A trial-and-error method to guess passwords or encryption keys by systematically checking all possibilities.
- **Botnet:** A network of compromised devices controlled by an attacker, often used for large-scale attacks like Distributed Denial of Service (DDoS).
- **Buffer Overflow:** An attack where extra data overflows into adjacent memory, potentially allowing attackers to execute arbitrary code on a system.
- **Cryptography:** The practice of encoding data to protect it from unauthorized access. Includes encryption and decryption.
- **Cross-Site Scripting (XSS):** A vulnerability that allows attackers to inject malicious scripts into websites, affecting users' browsers.
- **Denial of Service (DoS):** An attack aimed at shutting down a system, making it inaccessible to its intended users. DDoS is a distributed version involving multiple systems.
- **Exploit:** Code or method that takes advantage of a vulnerability in a system or software.

KEY TERMINOLOGY IN CYBERSECURITY

- **Hashing:** A process of converting data into a fixed-size hash value. Common in password storage and data integrity verification.
- **Honeypot:** A decoy system or network designed to lure attackers, helping in studying their tactics and gathering intelligence.
- **Injection Attacks:** Attacks that inject malicious code into a vulnerable program, such as SQL Injection, which targets databases.
- **Keylogger:** Malware that records keystrokes, allowing attackers to capture sensitive information like passwords.
- **Malware:** Malicious software designed to harm or exploit systems. Includes viruses, worms, Trojans, spyware, ransomware, and adware.
- **Man-in-the-Middle (MitM) Attack:** An attack where an attacker secretly intercepts and potentially alters communication between two parties.
- **Phishing:** A technique used to deceive users into providing sensitive information by pretending to be a legitimate entity.
- **Ransomware:** Malware that encrypts files and demands payment to restore access to data.
- **Rootkit:** A collection of malicious software tools that grant unauthorized root-level access to a computer.
- **Sandboxing:** A security practice where programs are run in isolated environments to prevent malicious code from affecting the broader system.
- **Session Hijacking:** Taking control of a user session by stealing or predicting session tokens.
- **Social Engineering:** The art of manipulating people into revealing confidential information. Phishing is a common example.
- **Spoofing:** Disguising as a trusted entity to deceive users or systems, such as IP spoofing, email spoofing, or DNS spoofing.
- **Spyware:** Software that secretly monitors user activity and collects data, often for malicious purposes.

KEY TERMINOLOGY IN CYBERSECURITY

- **SQL Injection:** A code injection technique that exploits vulnerabilities in applications to manipulate databases.
- **Trojan Horse:** A type of malware disguised as legitimate software that can provide unauthorized access to systems.
- **Virtual Private Network (VPN):** An encrypted connection over the internet that hides a user's IP address and provides a secure connection.
- **Virus:** Malware that attaches itself to a legitimate program and spreads by executing malicious code.
- **Vulnerability:** A weakness in a system that can be exploited by an attacker to compromise security.
- **Whitelisting:** A security measure where only pre-approved software, IPs, or users are allowed access to a system.
- **Zero-Day:** A vulnerability that is unknown to the vendor and has not yet been patched, making it highly exploitable.
- **DNS Spoofing:** Also called DNS cache poisoning, it redirects traffic to a malicious IP by altering the DNS records.
- **Cross-Site Request Forgery (CSRF):** An attack that forces a user to execute unwanted actions on a web application in which they're authenticated.
- **Cyber Kill Chain:** A model describing the stages of a cyberattack, from reconnaissance to actions on objectives.
- **Threat Intelligence:** Gathering and analyzing data about potential threats to prevent or mitigate cyberattacks.
- **Insider Threat:** A threat originating from within the organization, often involving malicious insiders or negligent employees.
- **Data Exfiltration:** The unauthorized transfer of data from a system, often used in data breaches and espionage.
- **Digital Forensics:** The process of uncovering and interpreting electronic data for use in legal cases or security investigations.

KEY TERMINOLOGY IN CYBERSECURITY

- **Incident Response:** A structured approach for handling and mitigating the effects of security breaches or attacks.
- **Penetration Testing (Pen Test):** An authorized simulated attack on a system to evaluate its security posture.
- **Red Team/Blue Team:** A simulated attack and defense exercise, where the Red Team simulates attackers and the Blue Team defends.
- **Advanced Persistent Threat (APT):** A prolonged, targeted cyberattack that stays undetected for an extended period, typically from nation-state actors.
- **Patch Management:** The process of updating software to fix vulnerabilities and improve security.
- **Security Information and Event Management (SIEM):** Software solutions that collect and analyze security events in real time to detect threats.
- **Access Control List (ACL):** A set of rules used to control the incoming and outgoing traffic of a network resource.
- **Air-Gapping:** Physically isolating a computer or network to prevent unauthorized access.
- **Cyber Hygiene:** Best practices for ensuring the maintenance of good security posture, like regular software updates and password management.
- **Anomaly Detection:** The identification of unusual patterns in data that do not conform to expected behavior, often used in Intrusion Detection Systems (IDS).
- **Advanced Encryption Standard (AES):** A symmetric encryption algorithm used for securing data. Widely adopted for its efficiency and security.

There are a lot of Terminologies present in the Cyber Security but right now we have covered Important and Basics part of it.
But if you wanted to know more Advance Terminologies in Cyber Security then Click here.

TYPES OF HACKERS





White Hat Hacker

Ethical hackers working legally to protect systems.





Gray Hat Hacker

Hackers who may break rules but don't have malicious intentions. Sometimes, they find vulnerabilities and report them without permission.





Black Hat Hacker

Criminal hackers who break into systems for malicious purposes.





Script Kiddies

Less skilled hackers who use existing tools and scripts to cause disruptions without fully understanding the technology.

WHITE HAT HACKERS // ETHICAL HACKERS





01 **Who They Are:** White hat hackers are cybersecurity professionals working legally to secure networks, websites, and systems. Often, they are hired by companies or governments.

02 **What They Do:** They perform security assessments, vulnerability scans, and penetration tests to identify and fix weaknesses.

03 **Why They Matter:** By finding and fixing vulnerabilities before cybercriminals can exploit them, white hats help protect sensitive data, maintain trust, and support overall cyber safety.

GRAY HAT HACKERS // JUST FOR FUN HACKERS





01 **Who They Are:** Gray hats sit between ethical and unethical hacking. They may explore system vulnerabilities without malicious intent, often for curiosity or self-education, and occasionally report their findings to the organization.

02 **What They Do:** Gray hats might discover and document security flaws but often without permission. They may inform the organization of the issues they found, sometimes expecting a reward.

03 **Why They Matter:** Gray hats can contribute valuable insights but raise ethical concerns, as they bypass official permission and could disrupt operations.

BLACK HAT HACKERS // UNETHICAL HACKERS





01 **Who They Are:** Black hats are cybercriminals who gain unauthorized access to systems for illegal purposes, such as stealing data, spreading malware, or disrupting services.

02 **What They Do:** Their attacks include hacking bank accounts, data theft, ransomware attacks, and cyber espionage.

03 **Why They Matter:** Understanding their methods helps cybersecurity teams anticipate and defend against real-world threats.

SCRIPT KIDDIES // HALF KNOWLEDGE HACKERS





01 **Who They Are:** Script kiddies are inexperienced hackers who rely on pre-made tools or scripts to execute attacks, often without deep knowledge of how these tools work.

02 **What They Do:** They perform low-skill attacks like website defacement, Distributed Denial of Service (DDoS) attacks, or other small-scale disruptions.

03 **Why They Matter:** While generally less threatening, they can still cause significant inconvenience and damage to smaller systems. Their activity often motivates them to improve their skills, potentially leading to more skilled black-hat behavior later.

WHITE-BOX TESTING // INSIDE TESTING





01 **What It Is:** In white-box testing, the tester has complete access to the system's internal architecture, source code, and design. This "inside-out" approach allows detailed exploration.

02 **How It's Done:** With code access, testers analyze how each component of the system works and identify flaws within specific sections.

03 **Purpose:** White-box testing helps find issues like code bugs, authentication issues, and design flaws. It is typically used by internal security teams to thoroughly inspect and improve system integrity.

GRAY-BOX TESTING // HALF-KNOWLEDGE





01 **What It Is:** Gray-box testing combines elements of both white- and black-box testing. The tester has limited knowledge of the system, typically basic architecture or previous results.

02 **How It's Done:** Testers use both internal knowledge and external approaches to identify vulnerabilities while maintaining a semi-outsider's view.

03 **Purpose:** Gray-box testing balances efficiency and realism, giving insights into how well systems handle attacks that come from semi-trusted users or employees with limited access.

BLACK-BOX TESTING // OUTSIDE TESTING





01 **What It Is:** In black-box testing, the tester has no prior knowledge of the system's structure or inner workings. The goal is to simulate the perspective of an external attacker.

02 **How It's Done:** Testers interact with the system solely from the outside, analyzing how data flows in and out and identifying accessible vulnerabilities.

03 **Purpose:** Black-box testing replicates a real-world attack and helps organizations understand what weaknesses an outside hacker might exploit. It's especially useful for public-facing systems, like websites or apps.

INTRODUCTION TO VM AND TOOLS

What is a Virtual Machine (VM)?

A Virtual Machine (VM) is essentially a "computer within a computer." It allows you to run multiple operating systems on a single physical machine, creating a safe and isolated environment for testing and experimenting.





Why Are VMs Important in Cybersecurity?

Virtual Machines are a key tool in cybersecurity because they let us:

01 **Test Without Risk:** You can experiment with cybersecurity tools and test various hacking methods without endangering your main system.

02 **Simulate Real Environments:** VMs can simulate different operating systems and network setups, which is perfect for practicing ethical hacking.

03 **Easy Reset:** If something goes wrong, like a malware infection, you can simply reset the VM to its original state.

SETTING UP OUR LAB ENVIRONMENT

In this training, we'll use VMs to safely explore cybersecurity concepts. The two main tools for setting up VMs are:

01 **VirtualBox:** Free software that allows you to create and manage virtual machines.

02 **VMware:** Another popular option, available in free and paid versions, widely used in cybersecurity labs.



INTRODUCTION TO PENTESTING OS



During this training, we'll be using different types of operating systems designed for penetration testing:

01 **Kali Linux:** The go-to OS for ethical hackers, packed with tools for network and application testing.

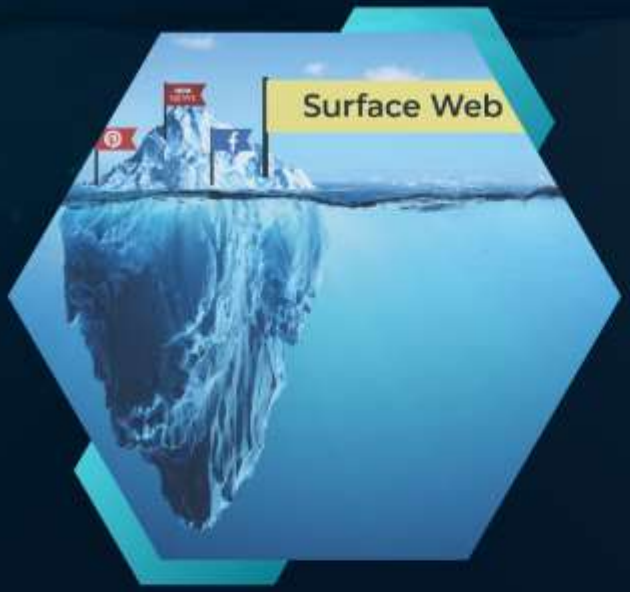
02 **Metasploitable 2:** A purposely vulnerable VM that's perfect for practice—it's designed for learning how to exploit weaknesses.

03 **OWASP Broken Web Applications:** A project offering a range of intentionally vulnerable web applications for web security testing.

04 **Parrot OS:** Similar to Kali, Parrot OS is also a penetration testing and forensics OS with various built-in tools.

05 **Windows 10 & Windows 7:** We use these VMs to understand how vulnerabilities affect common operating systems and to practice testing on familiar platforms.

TYPES OF WEB // SURFACE WEB



01 **What It Is:** The surface web is the part of the internet that is easily accessible and indexable by standard search engines like Google, Bing, and Yahoo.

02 **What's Found Here:** This includes any content visible to the general public, such as news websites, social media pages, blogs, and e-commerce sites.

03 **Usage:** When you perform a Google search, you're accessing the surface web. It's estimated to make up only about 4-5% of the entire internet.

04 **Example:** Websites like Wikipedia, YouTube, and CNN.

TYPES OF WEB // DEEP WEB



01 **What It Is:** The deep web refers to parts of the internet that are not indexed by standard search engines, making them less accessible to the average user.

02 **What's Found Here:** This includes private databases, internal company sites, academic journals, medical records, and other content requiring logins or permissions.

03 **Usage:** The deep web is used daily for secure and private access to sensitive information. Examples include online banking sites, email accounts, and private cloud storage.

04 **Example:** Your personal email inbox or your online banking dashboard. These are accessible but protected by login credentials.

TYPES OF WEB / DARK WEB



01 **What It Is:** The dark web is a part of the deep web but is deliberately hidden and requires special software, like Tor (The Onion Router), to access.

02 **What's Found Here:** The dark web hosts anonymous sites and is known for both legal and illegal content, including privacy-focused forums, secure communication platforms, and black markets.

03 **Usage:** It is used by individuals seeking privacy or anonymity, as well as by cybercriminals involved in illicit activities like drug trafficking, illegal firearms sales, and stolen data exchanges.

04 **Example:** Dark web marketplaces and forums require specialized tools and a deep understanding of anonymity practices to navigate safely.

SEARCH ENGINES

Standard Search Engines

01 **Examples:** Google, Bing, Yahoo.

02 **Functionality:** Index and display information from the surface web, making it easily accessible to the general public.

Specialized Search Engines for the Deep Web

01 **Examples:** Wolfram Alpha (for scientific queries), PubMed (for medical literature).

02 **Functionality:** Access specific databases and repositories within the deep web that are usually protected or restricted.

Dark Web Search Engines

01 **Examples:** Not Evil, Ahmia, TOR.

02 **Functionality:** Search engines designed to find content on the dark web, accessible only through networks like Tor. They prioritize privacy and security but require responsible use due to dark web risks.

Each web layer can be accessed using different types of search engines



Note: For **Academic Search Engines** you can go for **Google Scholar, JSTOR.**

SEARCH ENGINES

Why Understanding These Layers and Engines Matters in Cybersecurity

01 **Enhanced Research Abilities:** GKnowing where and how to access various information layers can improve threat intelligence gathering and research.

02 **Awareness of Hidden Threats:** Understanding the dark web can help professionals monitor and detect illegal activity that may affect organizations.

03 **Security Preparedness:** Knowing the layers helps cybersecurity experts secure and monitor the data on deeper web layers to prevent unauthorized access and breaches.



KEY TERMINOLOGY IN CYBERSECURITY (Extra)

- **Whaling:** A phishing attack targeted at high-profile individuals, such as executives, often with highly personalized messages.
- **Web Application Firewall (WAF):** A security tool that monitors and filters HTTP traffic to protect web applications from attacks.
- **Watering Hole Attack:** An attack strategy where hackers infect websites frequently visited by their target, hoping to compromise their systems.
- **Virtual Machine Escape:** An attack in which a malicious program in a virtual machine breaks out and gains access to the host system.
- **User and Entity Behavior Analytics (UEBA):** A security process that uses AI and machine learning to analyze the behavior of users and systems, identifying anomalies.
- **Tokenization:** The process of substituting sensitive data with a non-sensitive token to protect information in a secure manner.
- **Threat Hunting:** The proactive search for cyber threats within an organization's network to detect and mitigate risks before they cause harm.
- **Static Analysis:** The process of analyzing code for vulnerabilities without executing it, typically done by examining the source code.
- **Social Engineering Toolkit (SET):** An open-source framework used for simulating social engineering attacks for penetration testing purposes.
- **Sniffing:** The process of capturing and analyzing network packets to gather information, often used in network attacks.
- **Side-Channel Attack:** An attack that exploits physical signals, such as electromagnetic radiation, to extract data from a system.
- **Shoulder Surfing:** A social engineering tactic where attackers observe someone entering sensitive information, such as a password or PIN.
- **Security Operations Center (SOC):** A facility where security analysts monitor, detect, analyze, and respond to cybersecurity incidents in real time.

KEY TERMINOLOGY IN CYBERSECURITY (Extra)

- **Risk Assessment:** The process of identifying, evaluating, and prioritizing risks to protect systems and data effectively.
- **Rainbow Table:** A precomputed table used to crack hashed passwords by mapping hashes back to possible plaintexts.
- **Protocol Analyzer:** A tool used to capture and analyze network protocols, often used in network troubleshooting and security.
- **Pretexting:** A social engineering technique where attackers create a fabricated scenario to trick the victim into revealing information.
- **Pharming:** An attack that redirects users from a legitimate website to a malicious one, often by compromising DNS records.
- **Patch:** A software update that fixes vulnerabilities or bugs, often released by software vendors to improve security.
- **Passive Reconnaissance:** Gathering information about a target without direct interaction, typically through public sources or open data.
- **Packet Sniffing:** The act of capturing network traffic to analyze packets, often used for troubleshooting or malicious purposes.
- **Open Web Application Security Project (OWASP):** A nonprofit foundation focused on improving the security of software, known for the OWASP Top 10 list of common vulnerabilities.
- **Network Access Control (NAC):** A security solution that controls which devices can access a network, ensuring they meet security policies before connecting.
- **Multi-Factor Authentication (MFA):** A security system that requires more than one form of authentication to verify the user's identity.
- **Logic Bomb:** A malicious code that triggers a specific action when certain conditions are met, such as a specific date or event.
- **Least Privilege:** A security principle that restricts user access rights to the minimum necessary to perform their job functions.

KEY TERMINOLOGY IN CYBERSECURITY (Extra)

- **Intrusion Prevention System (IPS):** A network security device that monitors network traffic to actively prevent malicious activities or policy violations.
- **Information Security Management System (ISMS):** A set of policies and procedures for systematically managing an organization's sensitive data, ensuring confidentiality, integrity, and availability.
- **Indicator of Compromise (IoC):** Artifacts or evidence that indicate a system has been compromised, such as unusual file paths or IP addresses.
- **Identity and Access Management (IAM):** A framework that defines and manages the roles and access privileges of users in a network.
- **Heuristic Analysis:** A technique used to detect unknown viruses or malware by examining code behaviors instead of relying on known signatures.
- **File Integrity Monitoring (FIM):** A process that verifies the integrity of files to detect changes that may indicate a security incident.
- **Exploit Kit:** A set of tools or software designed to exploit vulnerabilities in systems, often sold on the black market for malicious purposes.
- **Endpoint Detection and Response (EDR):** A set of tools and solutions that monitor, detect, and respond to threats on endpoints such as computers and mobile devices.
- **Eavesdropping:** A passive attack where an attacker listens to network communications to steal information without altering the data.
- **Domain Generation Algorithm (DGA):** Algorithms that generate random domain names used by malware to evade detection by frequently changing C2 domains.
- **Digital Certificate:** An electronic document that authenticates the identity of a user or device, often used in SSL/TLS for secure web communications.
- **Deep Packet Inspection (DPI):** Analyzing the content of data packets passing through a network, often used in intrusion detection and filtering.
- **Data Loss Prevention (DLP):** Technologies and policies that prevent sensitive information from being lost, misused, or accessed by unauthorized users.

KEY TERMINOLOGY IN CYBERSECURITY (Extra)

- **Cyber Espionage:** The act of spying in cyberspace, typically used by nation-states to gain political, military, or economic advantages.
- **Common Vulnerabilities and Exposures (CVE):** A list of publicly disclosed information security vulnerabilities and exposures maintained by MITRE.
- **Command and Control (C2 or C&C) Server:** The server used by attackers to send commands to compromised devices or receive stolen data from a botnet.
- **Cloud Access Security Broker (CASB):** Security software that provides visibility, compliance, and threat protection for cloud applications and services.
- **Clickjacking:** An attack where a user is tricked into clicking on a concealed, malicious link, often through deceptive overlays.
- **Bot Herder:** The individual or group that controls a botnet, responsible for issuing commands to compromised devices.
- **Block Cipher:** A method of encrypting text in blocks rather than individual bits, often used in AES and DES encryption.
- **Big Data Security:** Protection measures for large volumes of data, ensuring data integrity, privacy, and protection against unauthorized access.
- **Authentication Factors:** The layers of proof required for authentication, typically grouped into:
  - Something you know (e.g., password).
  - Something you have (e.g., security token).
  - Something you are (e.g., fingerprint or biometrics).
- **Attack Vector:** The specific method or path an attacker uses to exploit a vulnerability, such as phishing, malware, or social engineering.
- **Attack Surface:** The total set of exploitable points or vectors an attacker can use to gain access to a system or network.
- **Asset Inventory:** A comprehensive list of all devices, software, and data within a network, crucial for identifying vulnerabilities and protecting resources.

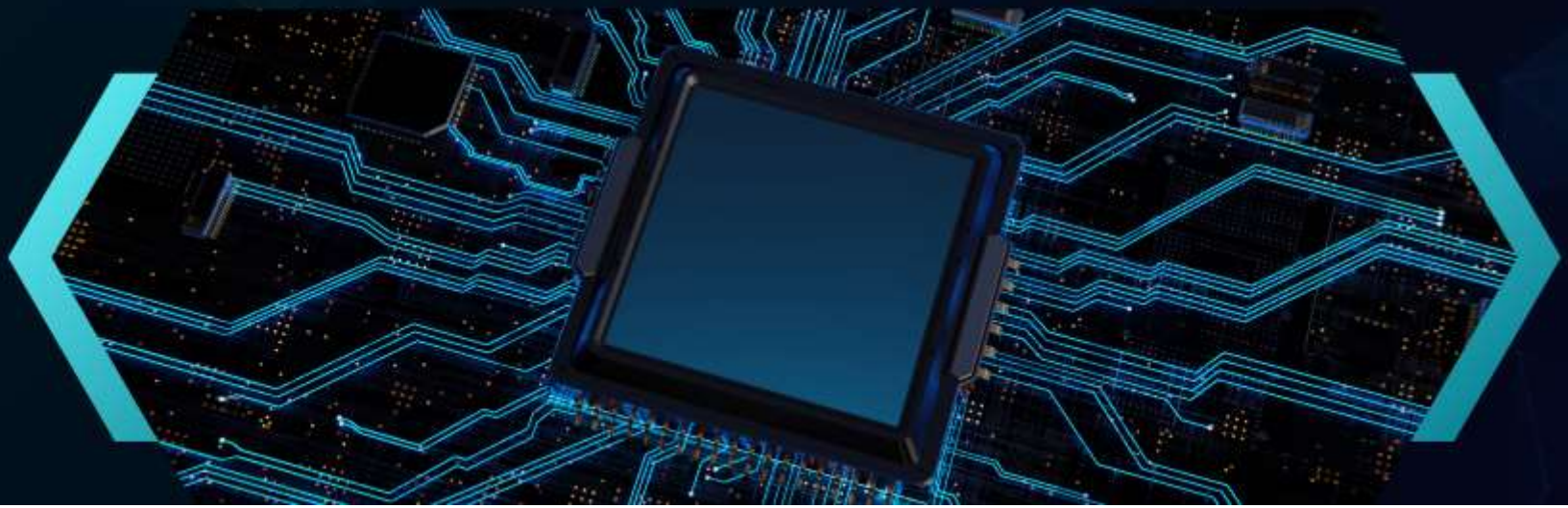
PRACTICAL EXERCISE 1

Basic Cyber Hygiene

Password Strength

01 **Goal:** Understand what makes a strong password. A secure password typically includes a mix of uppercase and lowercase letters, numbers, and special characters, avoiding easily guessable information.

02 **Exercise:** Compare passwords with different complexity levels using password strength checkers. Experiment with adding length and complexity to see how it affects security.



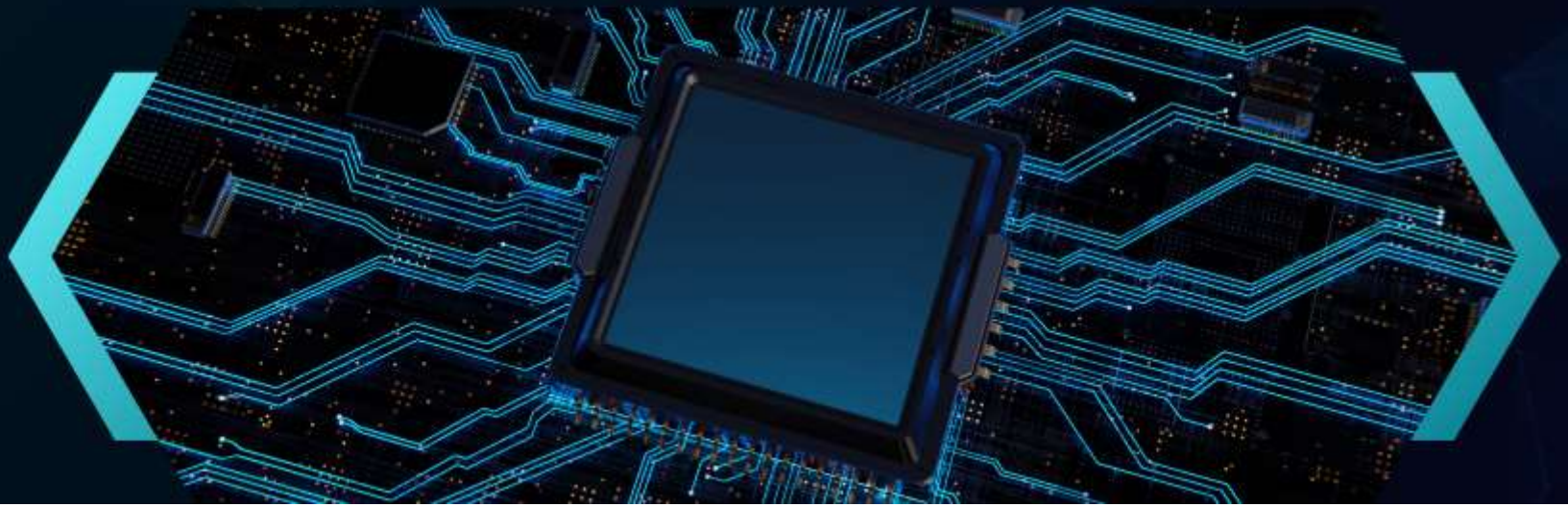
PRACTICAL EXERCISE 2

Basic Cyber Hygiene Part 2

Two-Factor Authentication (2FA)

01 **Goal:** Understand the added security of 2FA, which provides an extra layer of verification beyond a password.

02 **Exercise:** Set up two-factor authentication on accounts (e.g., email or social media) using apps like Google Authenticator or SMS. Discuss how 2FA can protect against unauthorized access even if a password is compromised.



PRACTICAL EXERCISE 3

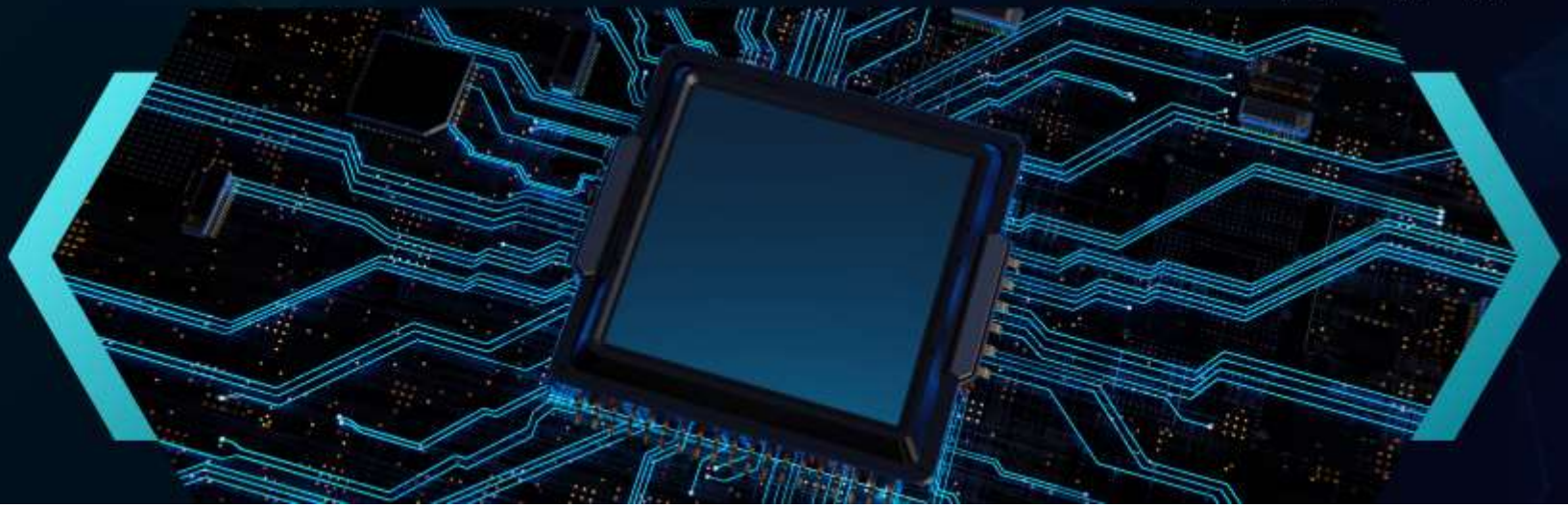
Malwares Creation

01 **Virus:** A virus is malicious code that attaches itself to a legitimate file, executing when the infected file is opened and potentially spreading to other files.

02 **Worms:** Worms are self-replicating malware that spread across networks without user interaction.

03 **Trojans:** A trojan appears as a legitimate program but performs malicious actions once executed.

04 **Ransomware:** It is a type of malware which prevents you from accessing your device and the data stored on it, usually by encrypting your files.



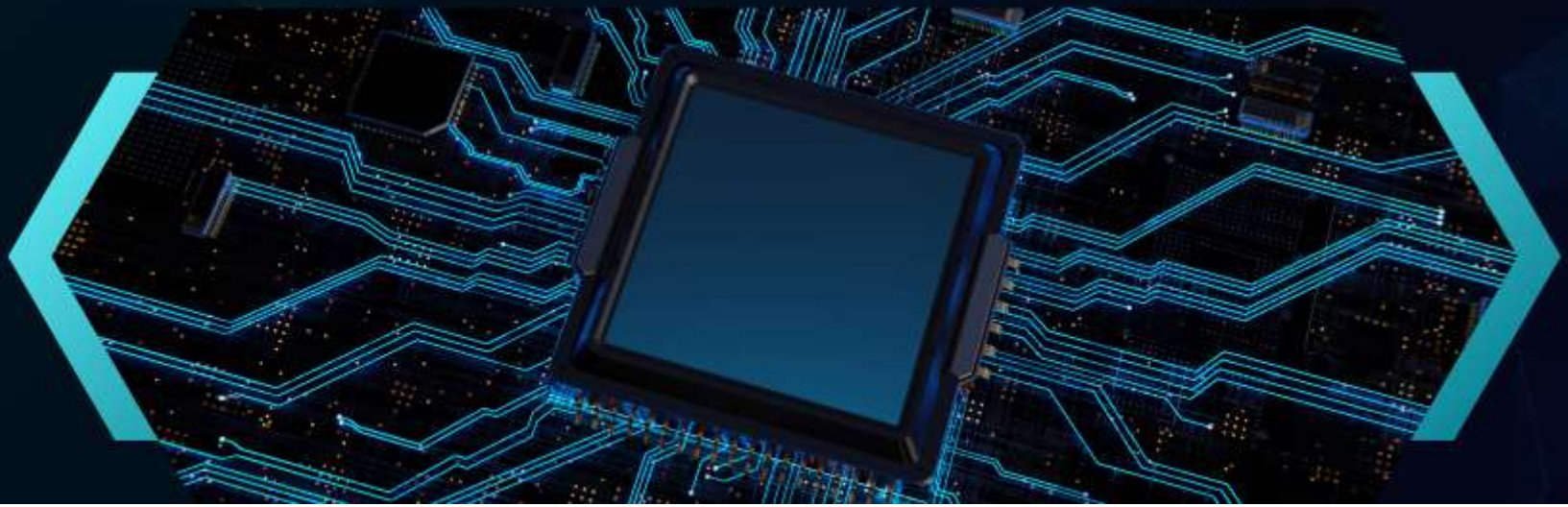
PRACTICAL EXERCISE 4

Setting Up a Lab Environment

VirtualBox/VMware Installation

01 **Purpose:** These tools allow users to run virtual machines (VMs) on their computer, isolating any testing or experiments from the host system.

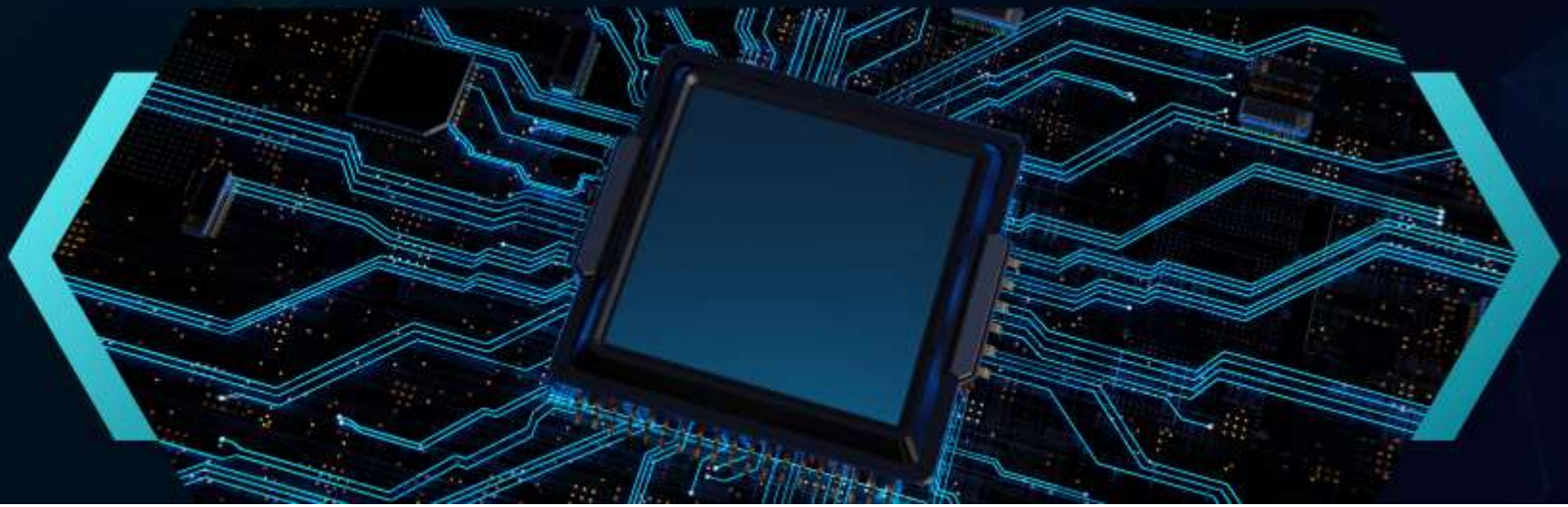
02 **Exercise:** Install VirtualBox or VMware on your computer. Practice creating virtual machines to run different operating systems, which will serve as safe environments for later exercises.



PRACTICAL EXERCISE 5

Installing Kali Linux and Another Penetration Testing OS

- **Install Kali Linux:** https://www.kali.org/get-kali/#kali-platforms
- **Install Metasploitable 2:** https://sourceforge.net/projects/metasploitable/
- **Install OWASP:** https://sourceforge.net/projects/owaspbwa/
- **Install Parrot OS:** https://parrotsec.org/
- **Install Windows 10 and 7**



PRACTICAL EXERCISE 6

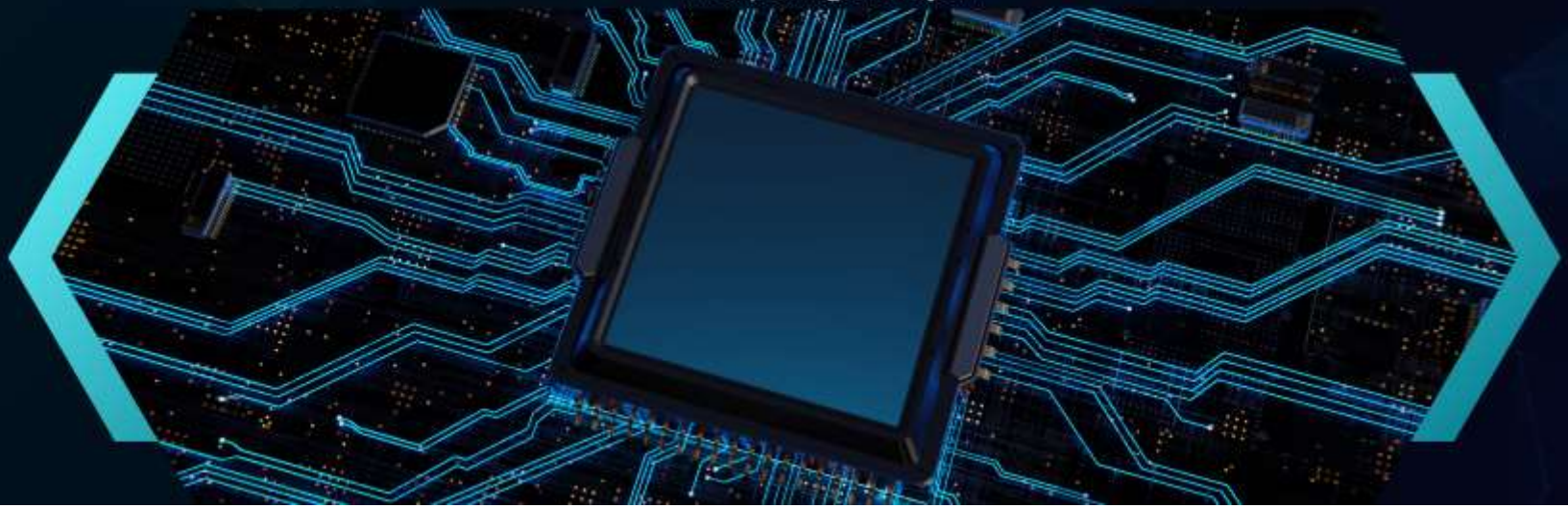
Accessing Types of Web

Identify and understand the Surface, Deep, and Dark Web

01 **Surface Web Access:** Explore common websites to recognize publicly accessible content.

02 **Deep Web Access:** Practice accessing parts of the web requiring logins or permissions (e.g., academic journals or cloud storage).

03 **Dark Web Overview:** Use Tor Browser to understand the dark web's structure and discuss risks and legal considerations involved in exploring this layer.





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